

```

        correct = tf.equal(tf.
argmax(prediction, 1), tf.argmax(y, 1))
        accuracy = tf.reduce_mean(tf.
cast(correct, 'float'))
        print('Accuracy:', accuracy.
eval({x:mnist.test.images, y:mnist.
test.labels}))

train_neural_network(x)

```

Now that we have that part done, let us take a look at the same implementation using *tf.keras* :

```

import tensorflow as tf
mnist = tf.keras.datasets.mnist

(x_train, y_train),(x_test, y_test) =
mnist.load_data()
x_train, x_test = x_train / 255.0, x_
test / 255.0

model = tf.keras.models.Sequential([
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(512,
activation=tf.nn.relu),
    tf.keras.layers.Dense(10,
activation=tf.nn.softmax)
])

```

```

model.compile(optimizer='adam',
               loss='sparse_
categorical_crossentropy',
               metrics=['accuracy'])
print("running...")
model.fit(x_train, y_train, epochs=5)
model.evaluate(x_test, y_test)

```

The Keras code is far simpler and the language is at a higher level than TensorFlow.

Colaboratory by Google

Increased performance in the execution of code is possible through the use of hardware like GPUs or TPUs rather than by running it solely on the CPUs. This is because CPUs, unlike the other two, have a limited number of cores and cannot process the vast mathematical calculations required even for quite a simple neural network. However, what can be done if your GPU is not supported by CUDA or you simply do not have one?

That is where services such as Colaboratory come into play. Colaboratory is an online service that lets you use an online notebook system called Jupyter to write and execute code which is wholly based on the cloud. This

avoids investing in expensive hardware such as a GPU to optimise code. Google Colaboratory gives you access to a Nvidia Tesla K80 GPU at no cost at all. The Colaboratory environment is exactly the same as any offline Python implementation. This means that there is absolutely no learning curve prior to starting off with Colaboratory. The fact that you run your code completely online means that you do not have to install any of the dependencies on your personal computer and this means that you are not restrained from implementing the latest ideas.

Putting your projects completely on the cloud allows you far more freedom and the access to Google TPUs which power the Colaboratory project. Unlike the Nvidia Tesla K80 GPU, the TPU does not come free, but you can use it for a basic fee calculated on TPU usage per second. This allows for the least possible execution time at the cheapest possible price. Like the TPU, Colaboratory also has several plans for other more powerful GPU prices, which are lower than the TPU rates, allowing you to choose what you feel is best for your particular use case.

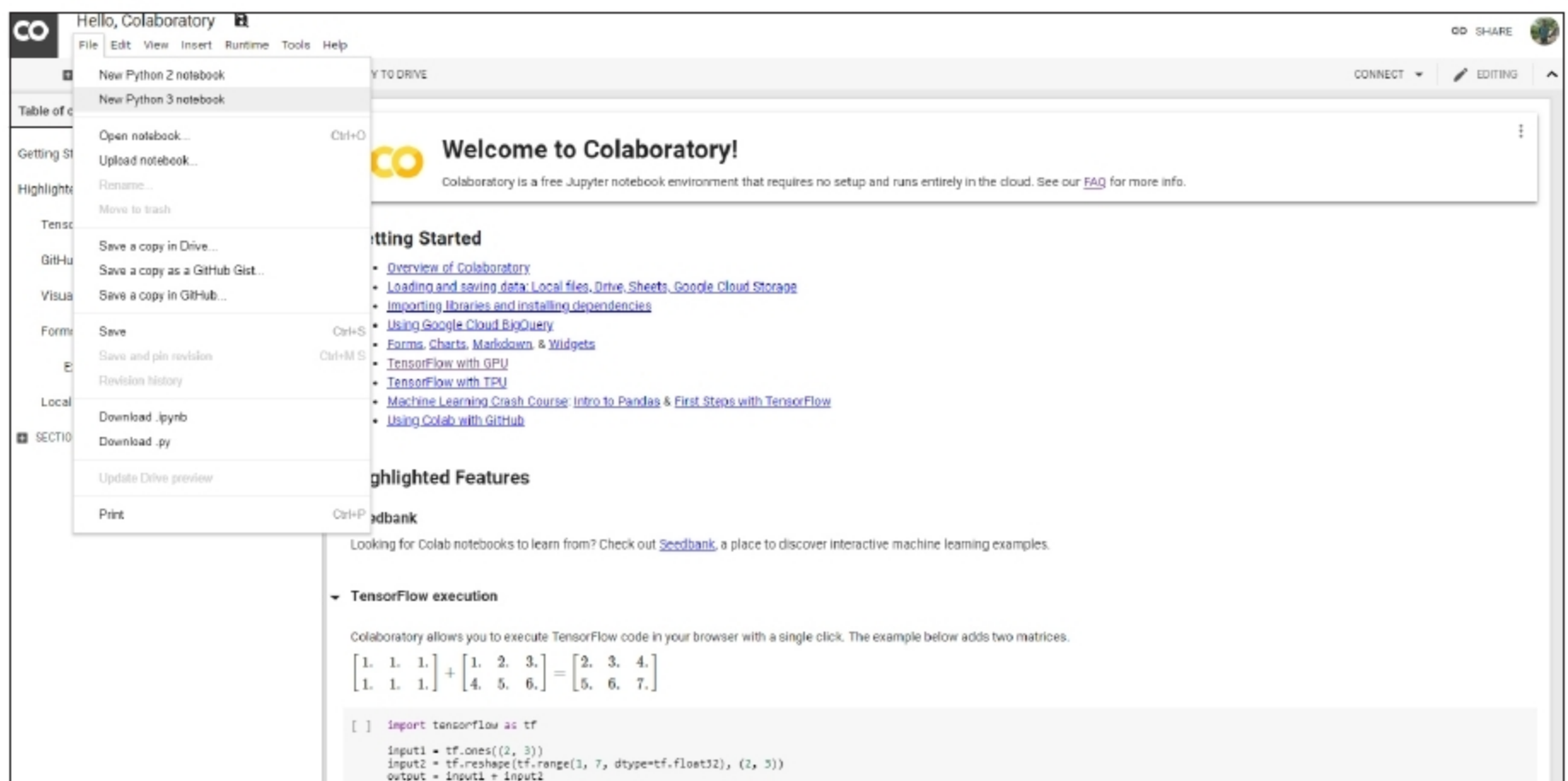


Figure 1: Creating a notebook